

definition 概率函数 Pr 的定义域是事件 (样本空间的子集):

$Pr[E] = \sum_{\omega \in E} Pr[\omega]$, 即任何事件的概率都是其中样本点概率的和.

$$\Rightarrow Pr[R=\tau] \stackrel{\text{def}}{=} \sum_{\omega: R(\omega)=\tau} Pr[\omega]$$

$\Rightarrow$ 断言 $R=\tau$ 本身即一个事件 $\Rightarrow$ 或者说, 任何关于随机变量取值的断言都定义了一个事件.

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$$[R=\tau] = \{\omega \in S \mid R(\omega)=\tau\}$$

$$\Rightarrow Pr[R=\tau]$$

18.2. Independence

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$$= Pr[\{\omega \mid R(\omega)=\tau\}]$$

Likewise $[M=1]$ is the event $\{TTT, HHH\}$ and has probability $1/4$.

More generally, any assertion about the values of random variables defines an event. For example, the assertion that $C \leq 1$ defines

$$= \sum_{\omega: R(\omega)=\tau} Pr[\omega]$$

$$[C \leq 1] = \{TTT, TTH, THT, HTT\},$$

$\downarrow$

and so $Pr[C \leq 1] = 1/2$.

一个随机变量的概率, 不是完全产生的, 而是从样本空间里按规则“筛”出来的. Another example is the assertion that $C \cdot M$ is an odd number. If you think about it for a minute, you'll realize that this is an obscure way of saying that all three coins came up heads, namely,

$$[C \cdot M \text{ is odd}] = \{HHH\}.$$

$\Rightarrow$ 概率在本质上一定是可数的, 即, 随机变量之所以能进行概率运算, 是因为它最终会落到可数样本空间可数的基本事件上.

18.2 Independence

$\Rightarrow$ 即使是在不可数样本空间上, 现代概率论也是在类似“筛”的思想上, 将任意随机事件分解为一组可数的子集类, 然后对每个子集类进行概率求和.

The notion of independence carries over from events to random variables as well. Random variables R_1 and R_2 are independent iff for all x_1, x_2 , the two events

对每个子集类进行概率求和.

$$[R_1 = x_1] \text{ and } [R_2 = x_2]$$

$\Rightarrow$ 即当看到 $Pr[R=\tau]$ 时, are independent.

其实是在计算, 属于同一类. For example, are C and M independent? Intuitively, the answer should be “no.” The number of heads, C , completely determines whether all three coins match; that is, whether $M = 1$. But, to verify this intuition, we must find some $x_1, x_2 \in \mathbb{R}$ such that:

的结果, 它们的原始概率

加起来是零.

$$Pr[C = x_1 \text{ AND } M = x_2] \neq Pr[C = x_1] \cdot Pr[M = x_2].$$

One appropriate choice of values is $x_1 = 2$ and $x_2 = 1$. In this case, we have:

$$Pr[C = 2 \text{ AND } M = 1] = 0 \neq \frac{1}{4} \cdot \frac{3}{8} = Pr[M = 1] \cdot Pr[C = 2].$$

The first probability is zero because we never have exactly two heads ($C = 2$) when all three coins match ($M = 1$). The other two probabilities were computed earlier.

On the other hand, let H_1 be the indicator variable for the event that the first flip is a Head, so

$$[H_1 = 1] = \{HHH, HTH, HHT, HTT\}.$$